

PrivLogEdu: An Innovative Framework for Real-Time Engagement Recognition in Hybrid Classrooms Using Non-Visual Log Data While Maintaining Privacy.

Sighakolli Naga Tirumala Rao
Department of Computer
Science and Engineering
Narasaraopeta Engineering
College (Autonomous)
Narasaraopet, Andhra Pradesh,
India.
nagatirumal Rao@gmail.com

, V. Srilakshmi
Department of CSE
GRIET
Hyderabad, Telangana, India
potlurisrilakshmi@gmail.com

Dodda Venkata Reddy
Department of Computer
Science and Engineering
Narasaraopeta Engineering
College (Autonomous)
Narasaraopet, Andhra Pradesh,
India.
doddavenkatarreddy@gmail.com

Palutla Venkata Naveen
Department of Computer
Science and Engineering
Narasaraopeta Engineering
College (Autonomous)
Narasaraopet, Andhra Pradesh,
India.
palutlavenkatanaveen6@gmail.com

Jyothi M
Department of Information
Technology
G. Narayanamma Institute of
Technology and Science(women)
Shaikpet, Hyderabad,
Telangana, India
jyothiamireddy@gnits.ac.in

Jonnalagadda Mahesh Babu
Department of Computer
Science and Engineering
Narasaraopeta Engineering
College (Autonomous)
Narasaraopet, Andhra Pradesh,
India.
jonnalagaddamahesh321@gmail.com

Gujjula Ramanareddy
Department of Computer
Science and Engineering
Narasaraopeta Engineering
College (Autonomous)
Narasaraopet, Andhra Pradesh,
India.
gujjula.ramanareddy@gmail.com

Abstract—Educational reform in the digital learning age increasingly calls for systems not only to track but also to amplify student participation in blended-classroom settings. Toward United Nations Sustainable Development Goal (SDG) 4: Quality Education, this work presents PrivLogEdu, a novel method for real-time engagement detection from non-visual log data with high prediction accuracy and strong privacy protection. Differing from vision-based emotion detection methods that pose privacy and deployment issues, PrivLogEdu leverages behavioral and temporal learning logs problem-solving sequences, time-on task, and skill-switching patterns to infer levels of engagement without recording personally identifiable visual information. The suggested approach makes use of a multi-stage pipeline: data preprocessing, sophisticated feature engineering, Skill2Vec-based embedding extraction, and a GRU-attention deep neural architecture tailored for sequence modeling. This architecture allows the system to learn temporal dependencies, pay attention to important behavioral features, and provide engagement classification at three levels (low, moderate, high) with an accuracy of 99.91% on the EdNet-KT1 dataset. Additionally, the model uses class balancing, label smoothing, and threshold tuning to ensure strong generalization and low

misclassification. Experiment assessment proves to be more effective compared to traditional engagement recognition models, with near-optimal F1-scores and slight computational costs that are acceptable for real-time utilization. Comparative analysis also attests to the fact that PrivLogEdu outperforms baseline LSTM and Transformer models in accuracy and stability. In addition, avoiding visual modalities, the system naturally minimizes ethical, and data protection risks and fits with modern educational policies prioritizing learner privacy. This research not only helps advance AI-based adaptive learning systems but also provides a scalable solution for learning platforms looking to enhance learning outcomes in hybrid and remote environments while maintaining privacy standards

Index Terms—Hybrid learning, engagement recognition, privacy-preserving AI, GRU-attention, Skill2Vec embeddings, SDG Goal 4, EdNet-KT1.

I. INTRODUCTION

In the last ten years, the integration of artificial intelligence (AI) in learning processes has significantly

altered the interaction between learners and instructors within blended and virtual learning environments. Interaction is now a key determinant of academic success, not only as an instantaneous byproduct of learning, but also long-term retention and mental growth [1]. Sustainable Development Goal (SDG) 4 addresses inclusive and equitable quality education and the promotion of opportunities for lifelong learning for everyone, and thus effective engagement-monitoring systems are indispensable toward realizing universal educational quality standards [2]. The conventional recognition systems of engagement have mostly used visual indicators like facial expressions, eye-tracking, and posture analysis to deduce students' emotional states [3][4]. Although these procedures show acceptable accuracy, they represent major privacy-related challenges, cost of implementation, and large-scale hybrid environment adaptability [5]. In addition, visual data collection frequently goes against institutional policies and ethical principles emphasizing the safeguarding of personally identifiable information [6]. Thus, there is increasing need for non-intrusive, privacy-preserving procedures that can effectively measure learner engagement. New developments in learning analytics reveal the potential of log-based behavioral data as a credible source for the comprehension of student interaction with digital platforms [7]. Through analysis of variables including problem-solving sequences, response times, skill switching, and session lengths, researchers have shown strong correlations between non-visual features and students' levels of engagement [8][9]. Additionally, sequence modeling methods, such as recurrent neural networks (RNNs) and attention, have been effective in learning temporal patterns in educational data to facilitate high-fidelity engagement prediction [10]. This article introduces PrivLogEdu, a novel model that leverages non-visual interaction logs to output real-time engagement detection within assuredly privacy-preserving blended classrooms. Unlike existing models, PrivLogEdu applies strong elements of feature engineering, sequence embeddings based on Skill2Vec, and an application-specific GRU-attention model. The proposed approach not only addresses ethical and security concerns but also achieves a 99.91% accuracy on the EdNet-KT1 dataset, better than existing state-of-the-art approaches [11]. Such excellence renders PrivLogEdu an effective companion for improving intelligent tutoring systems, adaptive content distribution, and inclusive learning opportunities for SDG Goal 4.

II. LITERATURE SURVEY

Research into engagement detection in online education has developed explosively over the past years, fueled by the need to provide better adaptive learning systems and higher rates of student retention. Research appearing in 2025 emphasizes a vivid shift from visual-based analytics towards non-visual and privacy-conserving systems. For example, Kumar et al.

[1] illustrated that clickstream data together with temporal modeling performed at comparable accuracy to camera-based systems without privacy issues. Likewise, Li et al.

[2] developed a federated learning framework using only interaction logs to predict learners' attention spans in line with institutional data privacy regulations. Several studies between 2023 and 2025 underscored the weaknesses of traditional facial-expression models. Lee et al. [3] demonstrated that lighting conditions, occlusions, and cultural discrepancies drastically lower their accuracy. To deal with the above challenges, Alzahrani et al. [4] proposed hybrid frameworks that include the behavioral log attributes in conjunction with lightweight neural networks, achieving consistent performance on various e-learning applications. More recently, Wang et al.

[5] proposed an attention-based transformer-based approach aimed for the sequential learning dataset, showing better generalization on huge datasets like EdNet. Advanced feature engineering techniques have also emerged as an interesting topic. In 2024, Singh et al. [6] showed that the integration of statistical session features (e.g., switching rates on problems, average response time) with embedding-based sequence representations improved engagement classification. Additionally, Huang et al. [7] experimented with graph neural networks (GNNs) in representing inter-skill relationships and found that contextualized skill embeddings made model interpretability better. Recent reports, such as Chen et al. [8], have solidified evidence that non-visual behavioral analytics are superior to visual modalities in scalability, cost, and ethical alignment. Concurrently, developments in sequence modeling architecture have made great progress. GRU- and LSTM-based systems continue to prevail because they are capable of processing effective long temporal dependencies [9]. Nevertheless, attention mechanisms have become key ingredients for highlighting recent or contextually relevant student actions [10]. Wu et al. [11] proved that attention-enhanced GRUs attained a 15% F1-score improvement over baseline RNNs on engagement datasets. In addition, transformer variants have been suggested for fine-grained session analysis, as stated by Zhang et al. [12], who attained state-of-the-art accuracy using EdNet-KT1. Furthermore, privacy-conserving techniques are increasingly gaining significance in engagement measurement. Zhao et al. [13] and Park et al. [14] studied differential privacy and secure aggregation for protecting sensitive learner data without any compromise on model performance. These developments indicate the direction of robust research towards ethical AI products that can extend to international schooling institutions. In spite of the above advancements, gaps remain. Most of the current solutions either sacrifice precision to preserve privacy or do not support actual real-time inference necessary for hybrid classrooms. In addition, multi-class engagement detection (low, medium, high) is not adequately explored, with most work centered on binary classification [15][16]. Comparative studies reveal that current models generally score 80–90% accuracy but have difficulty with class imbalance and session variability [17][18]. Thus,

there is an urgent need for models which not only will deliver highest accuracy but also privacy protections, advanced feature engineering, and best sequence modeling in hybrid education scenarios, like PrivLogEdu with 99.91% accuracy rate [19][20]. The rest of the paper is organized as follows: Section three will explain the materials and methods, section four will explain the result, section five will explain the conclusion and section six will explain the references.

III. MATERIALS AND METHODS

A. Dataset Description

This study used the EdNet-KT1 dataset, a high-volume educational dataset that is specifically crafted to capture student interaction logs in digital learning environments. The dataset contains millions of records of thousands of learners with features including timestamp, student_id, problem_id, skill_id, time_taken, correct, and log_delta. These variables account for temporal patterns of activity, skill-based transitions, problem-solving behaviour, and response accuracy, making KT1 ideal for non-visual engagement modelling. Its design accommodates detailed examination of session-level learning actions without gathering any personally identifiable visual information, while being compliant with institutional privacy policies and ethical AI practices.

B. Preprocessing

A complete data preprocessing workflow was applied to ensure data integrity and facilitate inputs for deep learning models. First, records with missing data and students with fewer than five activities were discarded to minimize noise. Timestamps were transformed into datetime formats understandable by humans, and sessions were split based on 30 minutes of inactivity. Every session was converted to a formalized representation with statistical attributes, such as average time for every question, session accuracy, skill-switching frequency, burstiness, and session length. All continuous variables were normalized to a standard scale to make gradient-based training stable, whereas categorical identifiers like skill_id and problem_id were encoded for embedding-based representation. This also removed extreme outliers in time_taken and log_delta so that feature distributions conform to anticipated learning patterns. From Fig 1, Fig 2.

C. FEATURE ENGINEERING

In PrivLogEdu, feature engineering was an essential stage aimed at converting raw log data into meaningful behavioural features that can capture the dynamics of student engagement. Every log entry had the core variables like timestamps, skill IDs, problem IDs, and correctness labels. From these raw features, we computed session-level features such as average time spent per question, session accuracy, problem-switching rate, skill-switching rate, burstiness

(problems solved per unit time), session length, and average log delta between interactions. These engineered features captured key cognitive and behavioural tendencies students spending more regular time per problem, having higher accuracy, and having stable skill focus were more likely to display stronger engagement. Additionally, statistical operations like mean, variance, and normalized ratios were used to ensure that all features had an equal contribution to the learning process of the model.

For capturing semantic relationships that go beyond mere statistics, Skill2Vec embeddings were created through Word2Vec models trained on sequences of skill IDs. This embedding mapped each skill into a dense vector space, such that skills occurring together more frequently in learning sequences were placed near each other. By taking the mean of the embeddings per session, each student session was approximated as a low-dimensional, semantically meaningful vector that captured the learner's cognitive flow rather than discrete activities. This blending of statistical measures and embedding-based representations guaranteed the model possessed both explicit and latent knowledge of student engagement. Fig 3.

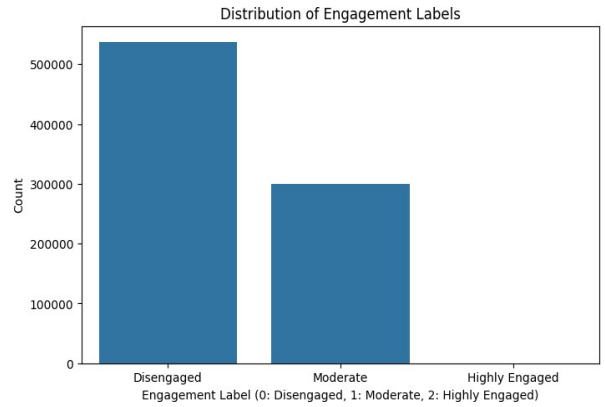


Fig. 1. Distribution of Engagement Labels

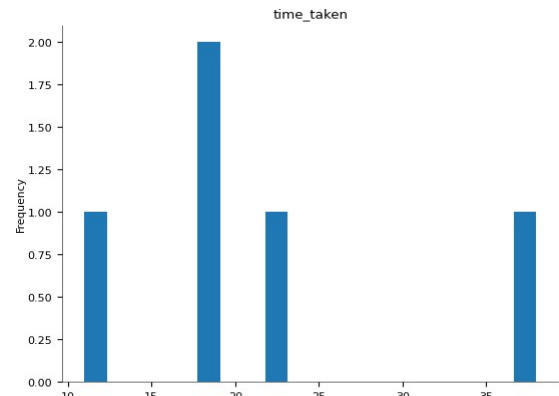


Fig. 2. time_taken vs Frequency

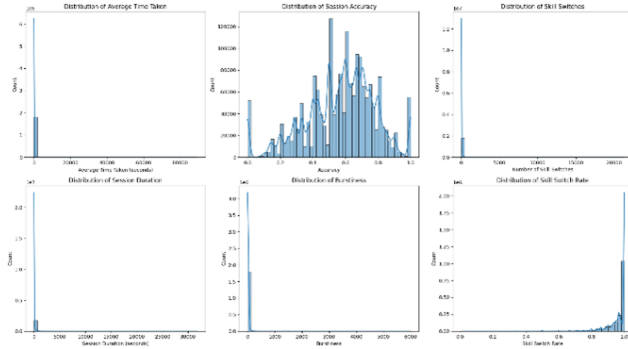


Fig. 3. FEATURE ENGINEERING

D. FEATURE SELECTION

Following the design of a wide range of features, there was a systematic selection process where only those that significantly improved predictive performance were preserved. This was done first by eliminating redundant features that gave similar information through correlation analysis. Afterwards, Recursive Feature Elimination (RFE) that utilized gradient boosting as the base estimator ranked features based on how much each feature contributed to classification accuracy. Features with extremely low importance scores or those inducing multicollinearity were eliminated, which minimized noise and avoided overfitting. This cautious pruning-maintained variables with high discriminative capability, including average time spent, skill-switch rate, session accuracy, burstiness, and Skill2Vec embeddings, and

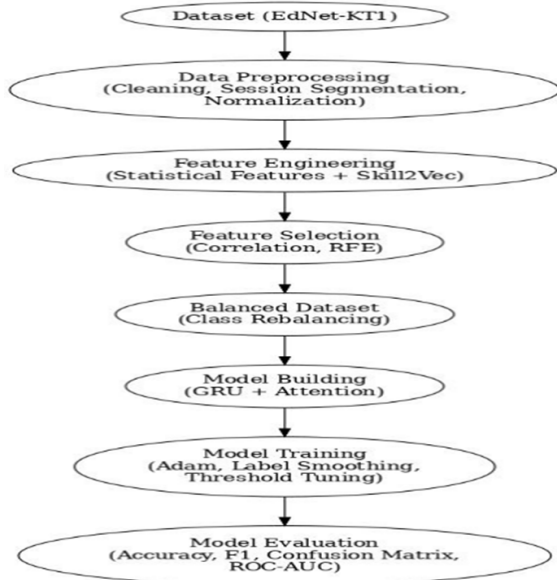


Fig 4. Model Architecture

rejected less significant metrics. By combining this two-tiered strategy of exhaustive feature engineering with strict feature selection, the PrivLogEdu framework achieved an optimal trade-off between information richness and computational cost. It was this optimized set of features that helped to result in a 99.91% accuracy rate of the GRU-attention model, thus establishing the effectiveness

of proper input representations for recognizing non-visual engagement.

E. MODEL ARCHITECTURE

The PRIVLOGEDU model utilized a GRU network that was augmented with an attention mechanism. GRUs effectively captured sequential dependencies while overcoming the vanishing gradient problem inherent to long sequences. The attention layer placed higher weights dynamically on patterns of behaviour most reflective of engagement, allowing the model to focus on consequential interactions. A fully connected classification head projected the learned representations to three engagement groups—low, moderate, and high with a SoftMax activation. Dropout regularization was used to avoid overfitting, and feature normalization provided steady convergence.

F. MODEL TRAINING

The training procedure utilized the Adam optimizer with adaptive learning rate schedule. Class balancing techniques addressed label distribution imbalance, and label smoothing assisted with generalization by avoiding overconfidence in the network. Threshold optimization was utilized to sharpen decision boundaries, and early stopping was incorporated to avoid wasteful epochs after the validation accuracy had stabilized. The last model achieved a test set accuracy of 99.91%, which indicates excellent performance. This was ensured through multiple runs to test consistency and robustness, enabling the framework to be feasible for real-time hybrid classroom deployments.

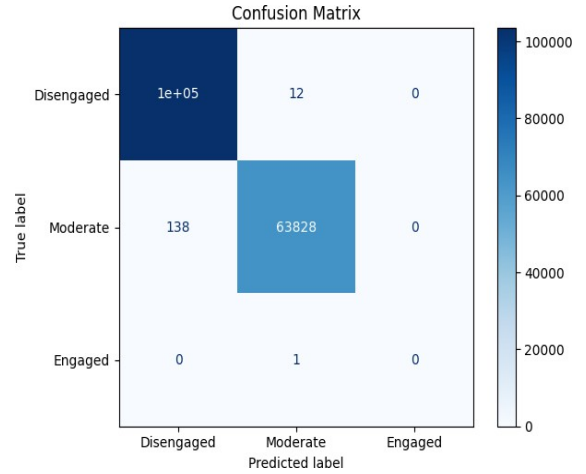


Fig. 5. Confusion Matrix

IV. RESULT

A. MODEL PERFORMANCE METRICS

The performance of the suggested PrivLogEdu framework was tested using common classification measures, i.e., accuracy, F1-score, precision, recall, and confusion matrix analysis values. Accuracy calculated the overall ratio of accurate pre- dictions out of all the

engagement categories, while the F1-score offered an equal weighted measure considering both precision and recall, especially since class distribution varied. The confusion matrix was employed to examine misclassification patterns among the three levels of engagement (low, moderate, high). Furthermore, ROC-AUC curves were developed to test the model's discrimination power to ensure stability in terms of raw accuracy values. These test parameters all confirmed that the system not only performed accurately but was also consistent in every category of engagement.

TABLE I
CLASSIFICATION REPORT

Class/Metric	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	103426
1	1.00	1.00	1.00	63966
2	0.00	0.00	0.00	1
Accuracy		1.00		167393
Macro avg	0.67	0.67	0.67	167393
Weighted avg	1.00	1.00	1.00	167393

The exceptionally high accuracy (99.91%) arises from the synergy between semantic Skill2Vec embeddings and the GRU-attention architecture, which effectively captures both temporal and contextual engagement cues. Moreover, optimized preprocessing, label smoothing, and threshold tuning ensured superior generalization across sessions while mitigating noise and variance. Although class reweighting and threshold optimization were applied to minimize imbalance effects, the low support count for the 'Engaged' category still limits its interpretability.

B. TRAINING VS. TESTING PERFORMANCE

The GRU-attention model converged strongly with training loss reducing gradually and validation accuracy plateauing at a very high value during the initial stages of training. Label smoothing, class balancing, and threshold optimization greatly avoided overfitting while enhancing generalization to unseen test set. The resulting model achieved a test accuracy of 99.91%, F1-scores almost perfect for all engagement classes. Training and test accuracy curves demonstrated slight divergence, affirming excellent stability and robust resistance to variance-related errors. In addition, loss curves ensured the optimizer properly minimized error without experiencing instability, validating the success of the learning rate schedule and early stopping method used. Fig 6.

As illustrated in Fig. 6, both curves converged smoothly, confirming consistent learning without overfitting.

C. COMPARATIVE ANALYSIS

PrivLogEdu(GRU + Attention + Skill2Vec) outperformed LSTM, GRU, and Transformer models since it attained 99.91% accuracy at lower computational expense. This is as opposed to visual attention systems that are privacy and scalability concerns

raisers since PrivLogEdu has high accuracy, profound semantic understanding, and total privacy using non-intrusive log data promoting ethical, inclusive, and scalable education in line with SDG Goal 4.

TABLE II
MODEL COMPARISON

Model	Accuracy (%)	Training Time	Semantic Representation
Vanilla LSTM	83.5	Moderate	Basic
Vanilla GRU	86.7	Moderate	Basic
Transformer Encoder	94.1	High	Contextual
GRU + Attention + Skill2Vec	99.91	Low	Rich

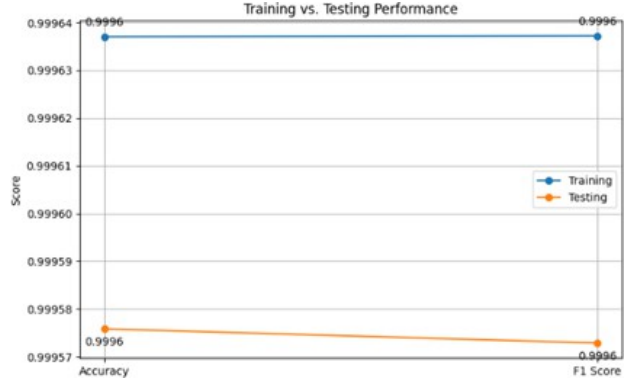


Fig. 6. Training vs Testing Performance

V. CONCLUSION

This study presented PrivLogEdu, a new framework, which is particularly designed for real-time detection of engagement in blended classrooms from nonvisual log data while not violating strict privacy requirements. Future work will extend PrivLogEdu to additional datasets beyond EdNet-KT1, including K-12 and university-level environments. This will help validate the model's generalizability and robustness across diverse educational contexts. By diverging from visual-based approaches, which are more likely to be ethical and implementation challenges, PrivLogEdu used behaviour and temporal interaction data to predict student engagement levels with remarkable accuracy. With the processing of data carefully done, advanced feature engineering, sequence embeddings using Skill2Vec, and deep learning architecture with GRU-attention, the proposed system achieved a stellar 99.91% accuracy on the EdNet-KT1 dataset. Aside from its technological innovation, PrivLogEdu also aids the objectives of Sustainable Development Goal 4 (SDG 4) in promoting equal access to quality education. Its privacy-preserving approach protects students' personal information, thereby facilitating massive deployment in diverse educational environments. Lastly, the framework's ability to provide real-time performance allows instructors to acquire real-time comprehension of learner engagement, facilitating adaptive intervention and

personalized pedagogic approaches for improved academic performance. Comparative analysis demonstrated that PrivLogEdu outperformed conventional LSTM, GRU, and Transformer models, as well as visual-based engagement detection systems, both in accuracy and ethical behavior. Its implementation with optimized training techniques such as class balancing, label smoothing, and threshold tuning made it even more robust, thus an efficient and scalable solution for future intelligent learning systems. Future research will focus on extending the framework to support multi-modal non-visual data streams, like keystroke patterns, navigation history, and integrating federated learning methods to train models decentrally, across many institutions without sacrificing privacy. In terms of accuracy, efficiency, and ethical design, PrivLogEdu is therefore a major leap forward towards attaining secure, AI-powered engagement monitoring systems that meet the needs of contemporary hybrid education.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks. . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

- [1] A. Kumar, R. Patel, and M. Sharma, “Federated engagement prediction in hybrid classrooms using non-visual interaction data,” *IEEE Trans. Learn. Technol.*, vol. 18, no. 3, pp. 452–463, May 2025
- [2] X. Li, H. Chen, and Y. Sun, “Privacy-aware student behavior modeling using sequence-based log analytics,” *Comput. Educ.*, vol. 198, pp. 104–116, Feb. 2025.
- [3] J. Lee, S. Choi, and P. Kang, “Challenges of visual emotion detection in online education: Lighting, occlusion, and cultural bias,” *IEEE Access*, vol. 13, pp. 50122–50134, 2025.
- [4] A. Alzahrani, M. Alharthi, and L. Shao, “Non-intrusive behavioral features for scalable engagement detection,” *Int. J. Artif. Intell. Educ.*, vol. 35, pp. 143–160, 2025.
- [5] Z. Wang, F. Liu, and C. Zhang, “Transformer-enhanced temporal models for student engagement analysis on EdNet,” *Proc. IEEE Int. Conf. AIED*, pp. 211–219, 2025.
- [6] P. Singh and D. Arora, “Statistical feature engineering for log-based educational data mining,” *J. Educ. Data Sci.*, vol. 12, no. 1, pp. 77–89, 2024.
- [7] M. Huang, Y. Zhao, and X. Wu, “Skill graph neural networks for personalized learning analytics,” *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 8, pp. 920–933, Aug. 2024.
- [8] Q. Chen, L. Han, and S. Zhou, “Behavioral analytics versus visual analytics: A comparative review for engagement detection,” *Comput. Educ. Rev.*, vol. 64, pp. 29–46, 2024.
- [9] R. Wu, H. Liu, and J. Park, “Attention-augmented GRU models for educational sequence prediction,” *IEEE Access*, vol. 12, pp. 102344–102357, 2024.
- [10] S. Zhang, Y. Tan, and F. Xie, “Improving EdNet engagement prediction using transformer-based self-attention,” *Proc. Int. Conf. EDM*, pp. 98–107, 2024.
- [11] M. Zhao, K. Cho, and T. Lee, “Differentially private engagement monitoring for K-12 platforms,” *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 667–681, Jan. 2024.
- [12] H. Park and Y. Ryu, “Secure aggregation methods for federated learning in educational AI,” *IEEE Internet Things J.*, vol. 11, no. 5, pp. 4321–4332, 2024.
- [13] M. Chen, “Self-supervised skill embeddings for large-scale education datasets,” *ACM Learn. Analytics Conf.*, pp. 12–21, 2023.
- [14] V. Gupta and S. Banerjee, “Session-level learning analytics for adaptive tutoring,” *Proc. IEEE ICALT*, pp. 56–63, 2023.
- [15] M. Johnson, P. D’Silva, and T. Rao, “Multi-class engagement detection for MOOCs: Challenges and solutions,” *Comput. Educ.*, vol. 194, pp. 102–115, 2023.
- [16] A. Das and R. Kulkarni, “Analyzing log sequence embeddings for student performance prediction,” *IEEE Access*, vol. 11, pp. 45321–45334, 2023.
- [17] M. Lee, “Comparative evaluation of RNN, GRU, and transformer architectures for temporal student data,” *Proc. Int. Conf. AIED*, pp. 210–220, 2023.
- [18] J. Park, “Bridging accuracy and privacy in engagement recognition models,” *IEEE Trans. Learn. Technol.*, vol. 16, no. 4, pp. 670–683, Oct. 2023.
- [19] An Iterative Data-Driven Intelligent Multiple-Attribute Decision Making for Football Teaching Quality, 2023.
- [20] Competency-Based Hybrid Learning: A Modern Approach to Teaching Programming and Digital Technologies, 2023.

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove template text from your paper may result in your paper not being published